

Why We Need Time Series Databases for Site Reliability Engineering

The recent explosion of IoT devices, an increased focus on site reliability engineering for hybrid clouds, real-time trading analysis, etc, have led to a greater spotlight on time series analysis and time series databases. This is because much larger time series data sets are being analysed today.



It's not uncommon to deal with petabytes of data today, even when carrying out traditional types of analysis and reporting. Traditional databases, however, do not offer optimal mechanisms to store and retrieve large scale time series data. To meet the demand of time series analysis, new types of databases are emerging.

Real-world use cases

Time series data streams are gathered in many real-world scenarios. The volume of data and the velocity of data generation differ from case to case.

Typical scenarios from different fields are described below.

- *Monitoring data gathered as part of site reliability engineering:* Health, performance, and capacity parameters are collected periodically over time from various layers of infrastructure and applications. These time series data streams are analysed for anomalies, fault detection, forecasting, etc. Huge amounts of time series data need to be analysed in real-time to avoid service breakdowns and for quick recovery. This data is also stored and retrieved for processing later, such as capacity forecasting.
- IoT devices and sensors generate continuous streams of data.
- Autonomous trading algorithms continuously collect data on how the markets are changing in order to optimise returns.
- The retail industry collects and monitors supply chain and inventory data to optimise costs.
- Weather forecasting teams continuously collect climate parameters such as temperature, humidity, etc, for predictions.